

Volume IV · Chapter 21 — RAG-vs-Fine-tune Paths & Synthetic Data · Labs

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-IV-Chapter-21-context.md` for the AI interaction log.

Prerequisites: Ch 11, Ch 12, Ch 20.

Learning outcomes — the student can: decide between RAG and fine-tuning for book knowledge; generate quality-controlled synthetic Q&A/instruction data from sources; build image-text pairs; compare accuracy/hallucination of both paths.

Labs (hands-on) — 16 h:

#	Lab	h
21a	Generate synthetic Q&A from a book chapter; QC filter	4
21b	Fine-tune a small model on the synthetic set (QLoRA)	5
21c	Build a RAG baseline over the same content	3
21d	Head-to-head: accuracy & hallucination, RAG vs. fine-tune	4

Datasets/tools: local LLM; HF `peft`/QLoRA; vector DB; eval harness. **Assessment:** comparative study (**60%**); synthetic-data QC report (**20%**); quiz (**20%**). **Key decisions:** RAG vs. fine-tune vs. both; synthetic-data quality gates; licensing of derived data. **References:** plan §6; §13 → *Fine-tuning...*; *RAG...* **Hours:** Theory **8** + Lab **16** = **24**.